

MEAN: make missing values explicit

Name the behavior you already know

01 / THE SAS EXAMPLE

```
data test;
  input x y z;
  average = mean(x, y, z);
  datalines;
1 2 3
4 . 6
. . .
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / ORIGINAL MEAN TRIALS

Original prompts; this edition adds a response-only instruction.
Model, client, settings and trial date: not stated in this PDF.

WHAT YOU WILL PRACTICE

Turn your SAS knowledge into instructions and a small checkable result.

Calculate one average per row. The three cases contain complete data, partly missing data, and all missing data. Keep all three when comparing the responses.

EXPECTED RESULT

Row	average	Reason
1	2.0	Three observed values
2	5.0	Missing value excluded
3	. / NaN	All values missing

SAS prints numeric missing as a period. The Python exercise uses NaN. A missing value is not zero.

Expected values checked in Python; SAS execution pending.

HISTORICAL TRIAL STATUS / SEE PAGES 2-3

A: 3/3 displayed matches | B2/B3: code reuse

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

ORIGINAL MEAN TRIALS / A1-A3

All three original A responses selected pandas and the same row-wise mean calculation. They displayed 2.0, 5.0, and NaN. Wording and code layout differed; A1 also offered a second version. These are displayed answers: the supplied A transcripts do not show the code being executed.

Historical evidence uses the original prompt. This edition adds the shared response-only instruction.

WHY CHOICES REMAIN OPEN

1 A clear destination

The request names Python. The SAS code supplies the calculation and example data.

2 Several open choices

The model chooses a library, data structure, missing-value representation, and output format.

3 Agreement can happen

A may already work. Record that honestly, then identify which decisions were inferred rather than stated.

ANTI-PATTERN TO AVOID

Anti-pattern: accepting the conversion because the complete row works. That checks only the easiest case. Check the partly missing and all-missing rows too.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas and NumPy.  
- Create a DataFrame with columns x, y, z in the given row order.  
- Represent SAS numeric missing values (.) with np.nan.  
- Calculate average across x, y, z within each row using  
  DataFrame.mean(axis=1, skipna=True).  
- Keep an all-missing row missing; do not replace missing with  
  zero.  
- Print x, y, z, average, and briefly explain the missing-value  
  rule.  
- Include assertions: averages are 2.0, 5.0, and NaN,  
  respectively.  
  Check NaN with pandas.isna(), not equality.
```

Show code in this response. Do not create or run files, or read existing Python files.

WHY THESE DETAILS HELP

- 1 Choose the tools**
Naming pandas and NumPy gives repeated answers a common structure.
- 2 State the rule**
Exclude missing values from each average; keep an entirely missing row missing.
- 3 Give a finish line**
Three expected results make the answer checkable. Run the assertions and review the results.

CHECK IT YOURSELF

Run the code. Confirm 2.0 and 5.0; check row 3 with pandas.isna(). Confirm three rows remain. If A also passes, record that result.

State the missing-value rule and include a row that proves whether the conversion follows it.

Original B2/B3 reused saved code; revised trials are pending.

SAS REFERENCE

One behavior, three implementations

One behavior contract, three implementations. Run the companion files and the same three acceptance tests against shared fixture data.

SAS / SOURCE BEHAVIOR

```
%macro translate(in=, out=);
data &out;
  set &in;
  average = mean(x, y, z);
run;
%mend;
```

SAS MEAN excludes numeric missing; an all-missing row remains missing. Loader maps `__MISSING__` to SAS `.` before translation.

PYTHON / PANDAS

```
def translate(df):
    out = df.copy(deep=True)
    cols = ["x", "y", "z"]
    out["average"] = out[cols].mean(
        axis=1, skipna=True)
    return out
```

pandas `mean(axis=1, skipna=True)` averages across named inputs; all-missing rows yield NaN. Loader decodes the numeric sentinel.

R / BASE

```
translate <- function(df) {
  out <- df
  cols <- c("x", "y", "z")
  a <- rowMeans(out[cols], na.rm=TRUE)
  a[is.nan(a)] <- NA_real_
  out$average <- a
  out
}
```

Base R `rowMeans(..., na.rm=TRUE)` returns NaN for an all-missing row; normalize it to `NA_real_`. Both are detected by `is.na()`.

Missing is not zero

The denominator is the number of observed values. A row with only one observed zero returns zero, while an all-missing row stays missing.

Preserve the inputs

The translation adds average to a new table. It retains x, y, z, row identifiers and row order, including their missing positions.

SAS MEAN pandas mean R rowMeans

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original three-row fixture

INPUT / CONDITION

Complete, partly missing and all-missing rows from the original example.

EXPECTED

average = [2, 5, missing].

WHAT THIS CATCHES

Wrong axis, propagating one missing input, or replacing all-missing with zero.

T2 / Added single-observation boundary

INPUT / CONDITION

Added validation rows [0, missing, missing] and [missing, -6, missing].

EXPECTED

average = [0, -6].

WHAT THIS CATCHES

Dividing by three, treating zero as missing, or excluding negative values.

T3 / Preservation, numeric type and order

INPUT / CONDITION

All shared rows in reverse order; non-default Python index.

EXPECTED

Original input values/missing masks retained; row order retained; average numeric.

WHAT THIS CATCHES

Input mutation, row loss/reordering, index reset, or text instead of numeric results.

REFERENCE CODE + TESTS: [module-01-mean/appendix/](#)

Run: `python tools/run_appendix_tests.py --modules 01`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.